MODULE 03

Data Cleaning & Preprocessing

Garbage In, Garbage Out — Fix It First

■ DATA QUALITY

■ LEARNING OUTCOME

By the end of this PDF, you will confidently identify and fix the most common data quality problems using Excel and Python, preparing any dataset for analysis.

■ Skills You Will Gain
• Identify missing, duplicate, and inconsistent data
• Handle missing values (drop, fill, impute)
• Remove duplicates and fix data types
• Normalize and standardize numeric data
• Encode categorical variables for analysis

■ **80% of a data analyst's time is spent cleaning data. Master this = master analytics.**

SECTION 1

The Dirty Data Problem

Problem Type	Example	Impact
Missing Values	Age = NaN, Salary = blank	Wrong averages, model failures
Duplicates	Same order recorded twice	Inflated totals, false counts
Wrong Data Type	Date stored as text '01-Jan'	Cannot sort or calculate
Inconsistent Format	'Male', 'male', 'M' in Gender	Wrong group counts
Outliers	Salary = ■99,99,99,999	Skewed mean, bad predictions
Irrelevant Columns	CustomerID, Timestamp for name analysis	Noise in model

SECTION 2

Handling Missing Values

1. **Detect**
   df.isnull().sum() — count missing per column
2. **Decide**
   Is it <5%? Drop rows. Is it >20%? Drop column or impute.
3. **Fill Numeric**
   df['col'].fillna(df['col'].mean()) — fill with mean
4. **Fill Categorical**
   df['col'].fillna(df['col'].mode()[0]) — fill with most frequent
5. **Advanced**
   Use KNNImputer or median for skewed data

Python Code:

```
import pandas as pd df = pd.read_csv('data.csv') print(df.isnull().sum()) # Check missing
df['Age'].fillna(df['Age'].median(), inplace=True) # Fill with median
df.dropna(subset=['Salary'], inplace=True) # Drop rows with missing Salary
```

SECTION 3

Removing Duplicates

```
df.duplicated().sum() # How many duplicates? df.drop_duplicates(inplace=True) # Remove all
duplicates df.drop_duplicates(subset=['OrderID'], keep='first') # By specific column
```

SECTION 4

Fixing Data Types

```
df.dtypes # Check all column types df['Date'] = pd.to_datetime(df['Date']) # Convert to date
df['Price'] = df['Price'].astype(float) # Convert to number df['Category'] =
df['Category'].astype('category') # Memory efficient
```

SECTION 5

Standardization & Normalization

Technique	Formula	When to Use
Min-Max Normalization	(x - min) / (max - min) → [0,1]	When you need values between 0 and 1
Z-Score Standardization	(x - mean) / std → mean=0, std=1	When data is normally distributed
Log Transform	log(x) — reduces skewness	Skewed data like salary, price

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler scaler = MinMaxScaler()
df[['Age','Salary']] = scaler.fit_transform(df[['Age','Salary']])
```

SECTION 6

Encoding Categorical Variables

```
# Label Encoding (for ordinal: Low, Medium, High) from sklearn.preprocessing import
LabelEncoder le = LabelEncoder() df['Grade'] = le.fit_transform(df['Grade']) # One-Hot
Encoding (for nominal: Red, Blue, Green) df = pd.get_dummies(df, columns=['Color'],
drop_first=True)
```

■ PRACTICE QUESTIONS

Q1. What function in Python shows how many missing values are in each column?

Answer: *df.isnull().sum()*

Q2. When should you use median instead of mean to fill missing values?

Answer: *When the data is skewed (has outliers) — median is not affected by extreme values.*

Q3. What is the difference between normalization and standardization?

Answer: *Normalization scales to [0,1]; standardization scales to mean=0, std=1.*

Q4. Why should you remove duplicate rows before analysis?

Answer: *Duplicates inflate counts and totals, leading to incorrect analysis results.*

Q5. What is one-hot encoding and when do you use it?

Answer: *Converting categorical variables into binary columns. Use for nominal (unordered) categories.*

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: employee_dirty.csv (Name, Age, Salary, Department, Join_Date)

Steps:

1	**Detect** Run df.isnull().sum() and df.duplicated().sum()
2	**Fix Missing** Fill Age with median, Department with mode
3	**Remove Dups** Drop duplicate rows based on Name + Join_Date
4	**Fix Types** Convert Join_Date to datetime, Salary to float
5	**Normalize** Apply Min-Max scaling to Age and Salary
6	**Encode** One-hot encode Department column

Expected Output: A clean, analysis-ready DataFrame with no nulls or duplicates

Insights to Find: How many rows were removed? What % were dirty? Distribution of departments.

■ Pandas Data Cleaning Cheat Sheet	
`df.shape`	`Rows and columns count`
`df.info()`	`Data types + null counts`
`df.describe()`	`Stats: mean, std, min, max`
`df.isnull().sum()`	`Missing values per column`
`df.duplicated().sum()`	`Count duplicate rows`
`df.drop_duplicates()`	`Remove duplicate rows`
`df.fillna(value)`	`Fill missing with value`
`df.dropna()`	`Remove rows with any null`
`df.astype(type)`	`Change column data type`
`pd.to_datetime(col)`	`Convert text to datetime`
`df.rename(columns={})`	`Rename columns`
`df.drop(columns=[])`	`Remove columns`